

Automated Workstation Reset.

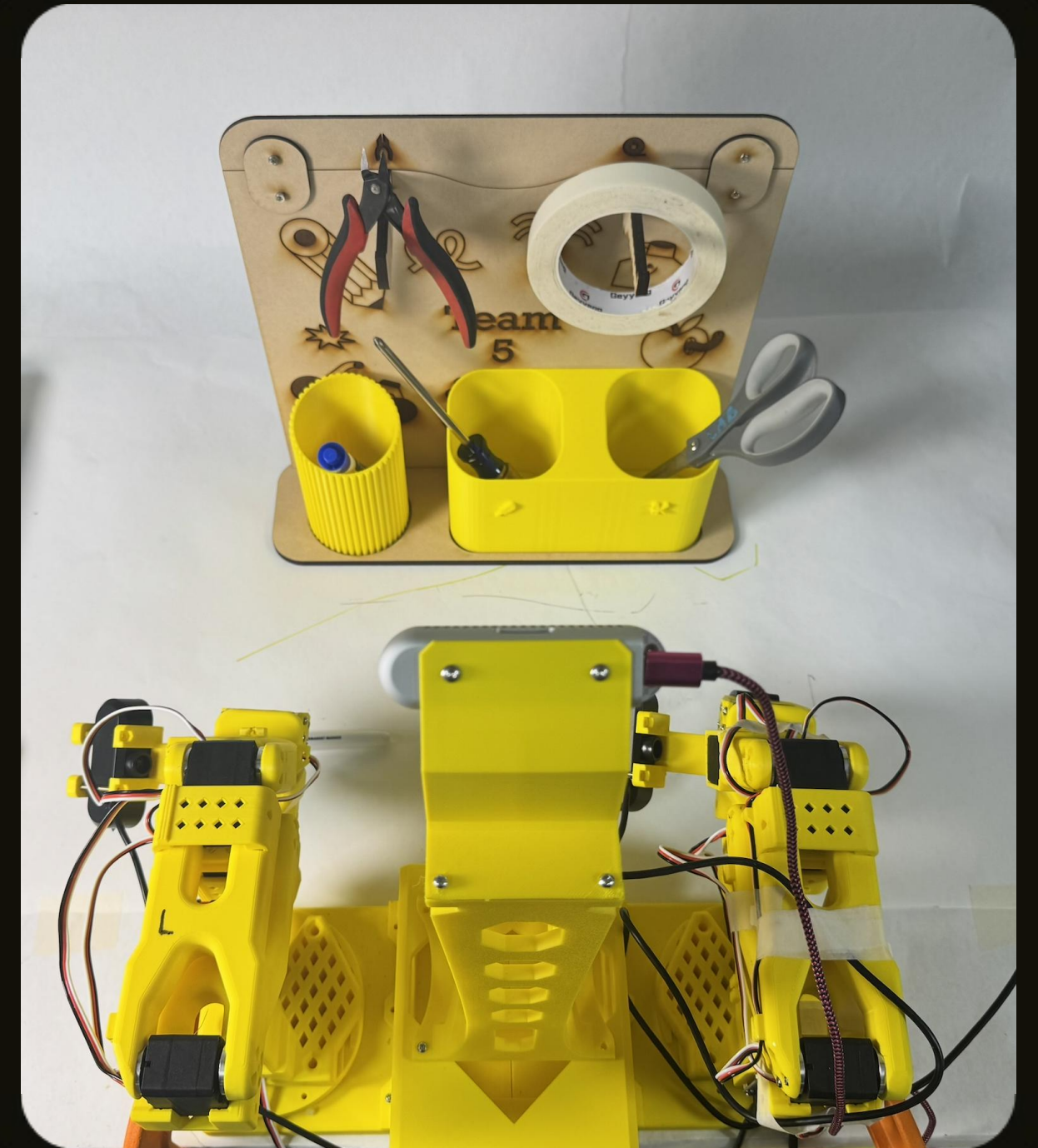
A dual-arm robotic system that detects, sorts, and returns tools to their designated workstations.

TEAM

Suzy Hong · Chenming Ge · Lya Liu

DATE

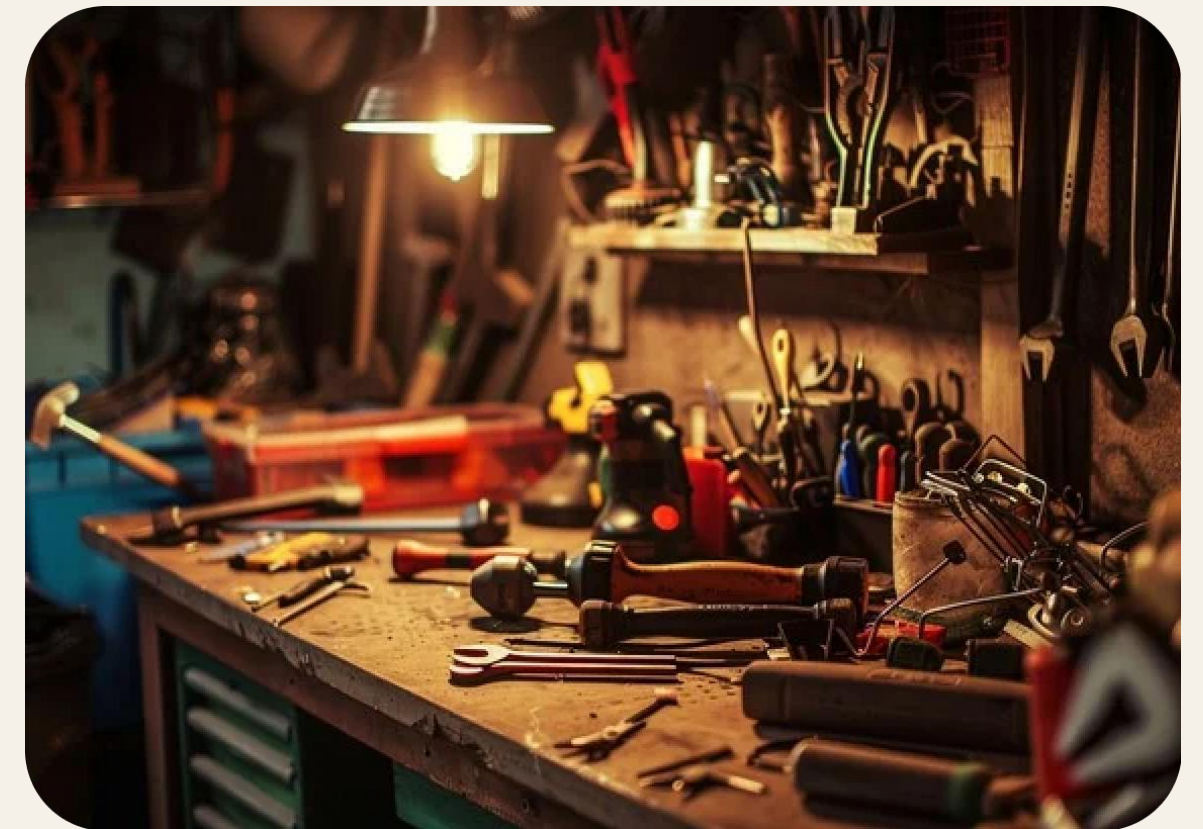
Jun 8 2026



Makerspaces get messy. Resetting them is the boring part.

Industrial workshops rely on 5S: Sort, Set in order, Shine, Standardize, and Sustain. This method keeps workspaces ready for the next person.

Our bimanual SO-101 automates a repetitive reset task: detecting loose tools and returning them to the right place.



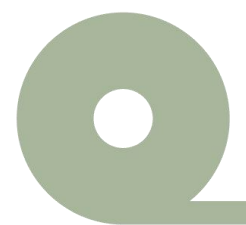
THE FIVE TOOLS WE HANDLE



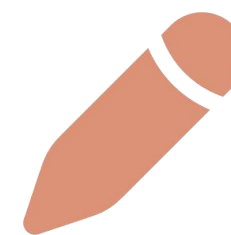
SCREWDRIVER



PLIER



TAPE

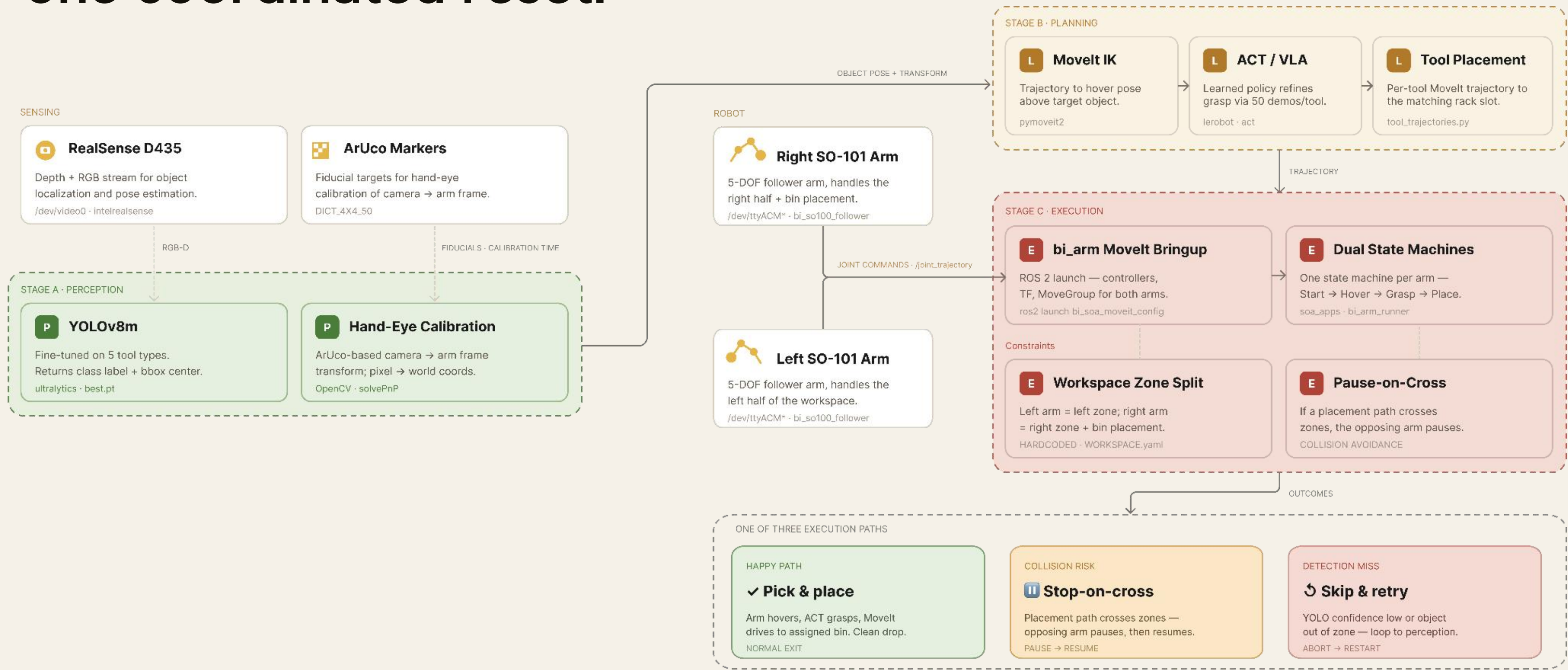


PEN



SCISSOR

Three stages, two arms, one coordinated reset.





plier ✕
scissor ✓

What broke, and what we changed.

WHAT ISSUES		WHAT WE CHANGED		OUTCOME
Camera calibration kept drifting	→	Swept multiple methods and traced root cause to ArUco cube face distortion	→	Stable pixel → world transform for YOLO
★ Single arm ran fine; dual-arm launch broke on TF + parameters	→	Many rounds patching the launch file, eg. namespaces, TF prefixes, controller configs	→	Just-functional dual-arm bringup; not yet stable
YOLO kept misclassifying background objects as the target tool	→	Two extra fine-tune rounds + parked the arm outside the RealSense view at start	→	Cleaner detections; false positives largely gone for condition A & B
5-DOF reach + invalid start states broke planning	→	Position-only IK fallback + relaxed start poses	→	Reliable planning on real hardware

How we measured success.

SUCCESS DEFINITION · BINARY

PASS: Object is correctly placed in the designated box and completes the full pipeline.

FAIL: Wrong box, dropped, or timeout (>2min)

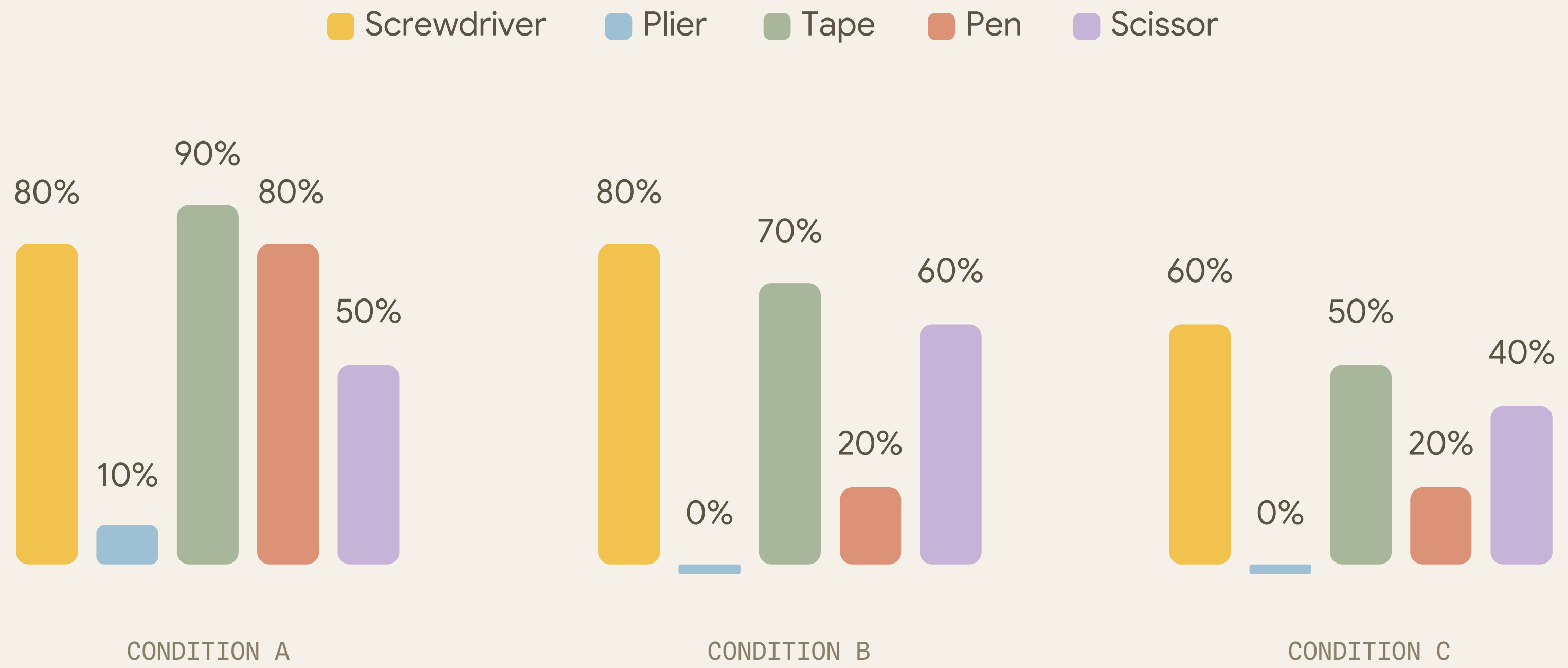
THREE CONDITIONS · 10 TRIALS EACH · 30+ RUNS

- A** 1–2 identical objects, randomly placed
- B** 3 mixed objects, randomly placed
- C** 3 random objects + clutter — paper scraps, debris

METRICS RECORDED

SUCCESS RATE Percentage of successful completions per object type per condition (target: >90%)	CYCLE TIME Total time from object detection to successful placement (seconds) (Mean ± std)	PER-STAGE Hover: Front camera verification (≥50% object visibility) Pickup: Correct grasp region secured Trajectory: Accurate placement in target box	CONSISTENCY Standard deviation of cycle time across trials. Lower values indicate more predictable and reliable performance
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Success rate by condition and tool.

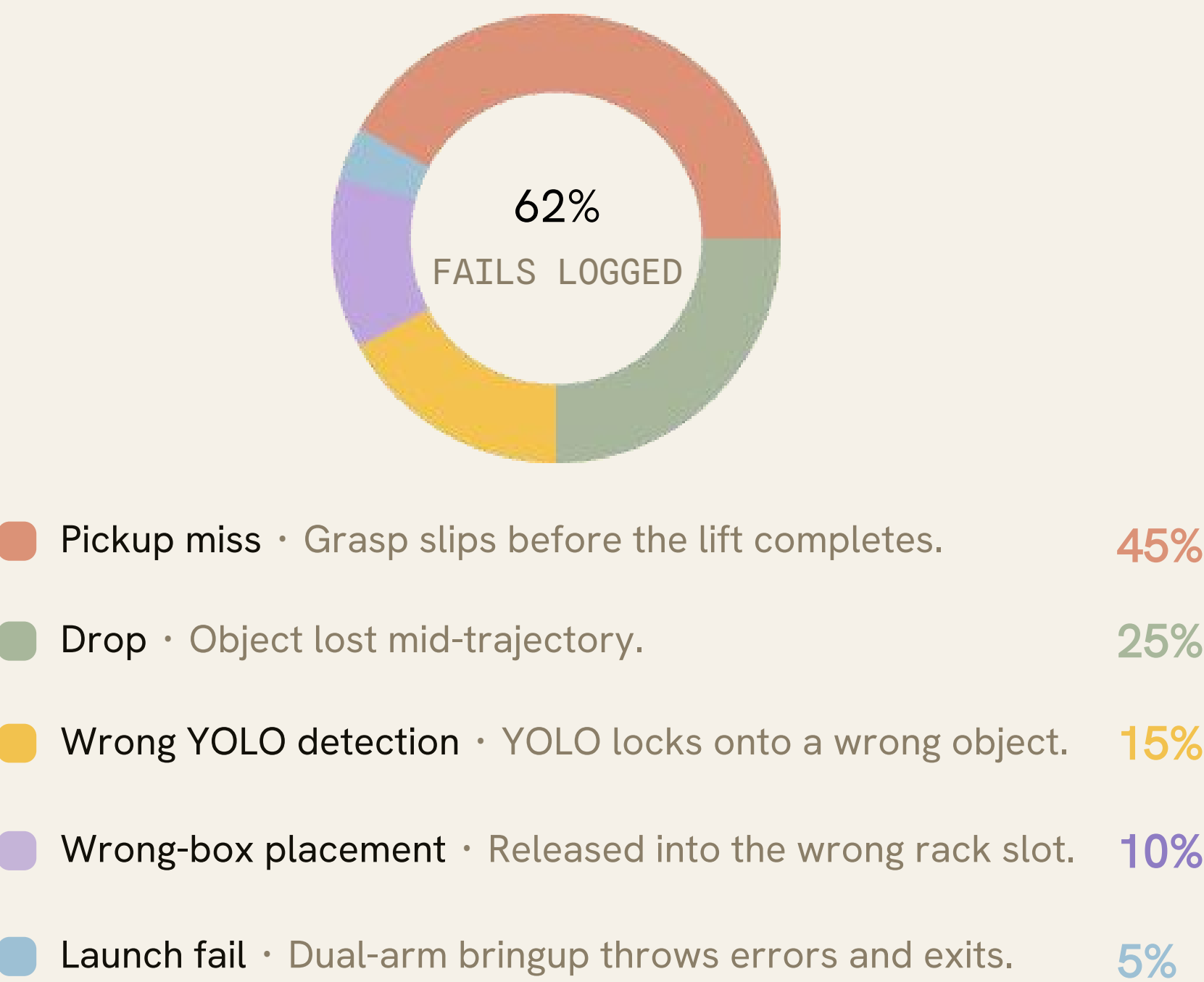


How fast, and how it fails.

CYCLE TIME · MEAN ± STD (S)

STAGE	COND A	COND B	COND C
Hover	7~10	9~12	10~14
Pickup	20	20	20
Placement	20	20	20
Full Trial	47~90	95~115	95~135

FAILURE MODES · CATEGORIZED PER CYCLE (total = 130)



Beyond raw success rate.

■ Safety

Across all trials, we recorded zero arm-to-arm collisions and no unsafe gripper-to-table contact. The hardcoded workspace zoning we introduced after Pre1 holds up reliably.

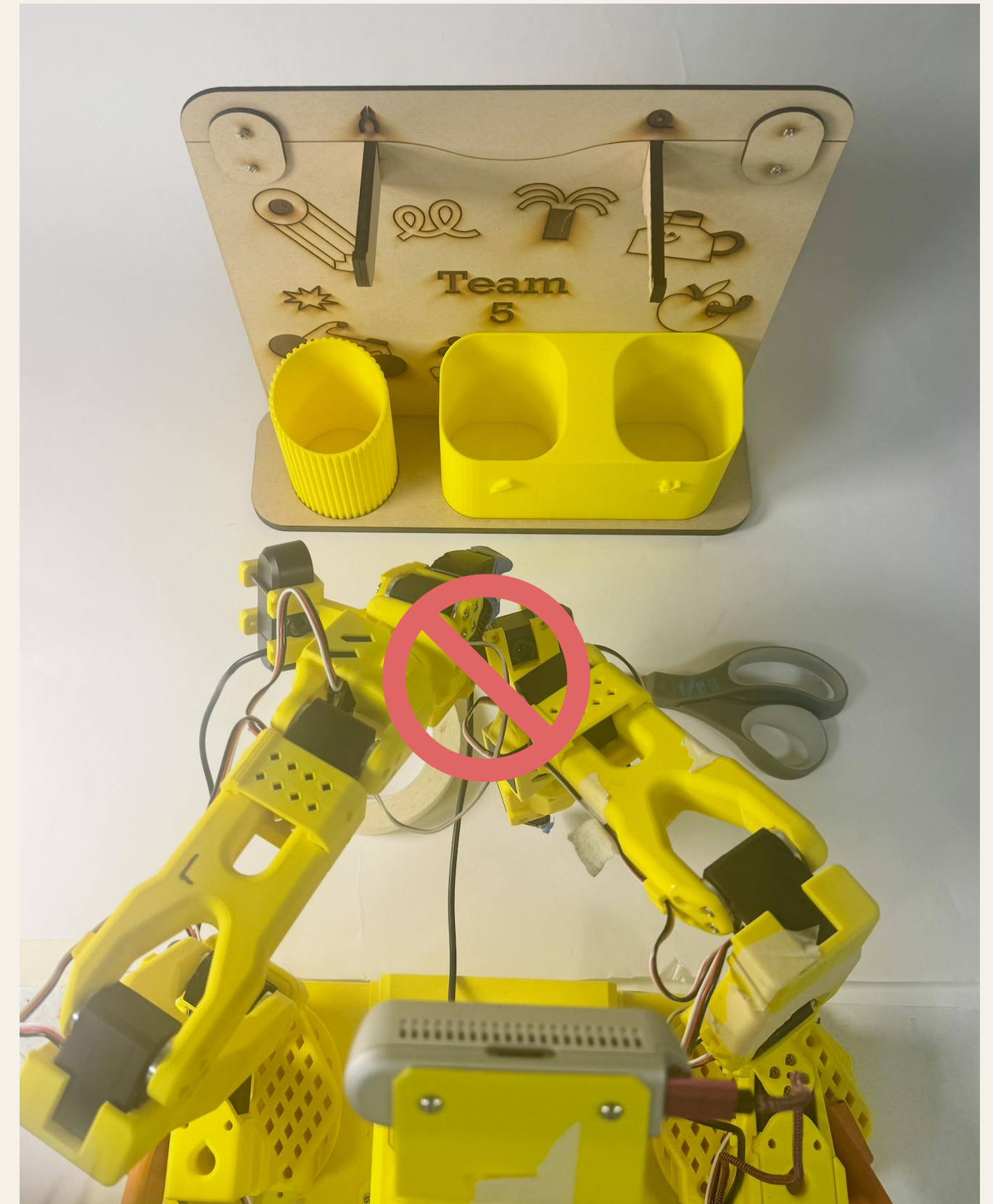
■ Robustness

The system runs 4 consecutive cycles before needing a reset, recovers gracefully when YOLO misses a detection, and stays stable across repeated runs.

■ Generalization

We tested how the policy performs on related but unseen tools, under varied lighting, and with cluttered backgrounds.

This tells us whether the model is truly learning manipulation, or just memorizing our setup.



AUTOMATED WORKSTATION RESET.

Thank you.

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